Third Normal Form (3NF) Relational Database Design

> **Definition:** A relation is in **Third Normal Form (3NF)** if it is in **2NF** and **no non-prime attribute is transitively dependent on the primary key**. For every non-trivial functional dependency X -> Y, either **X is a super key** or **Y is a prime attribute**.

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1. Detailed Technical Explanation

Transitive Dependency Concept:

A transitive dependency exists if A -> B and B -> C hold, which implies A -> C through non-key attribute B.

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### Conversion Example: 2NF to 3NF

#### Relation EMP_DEPT(Emp_ID, Emp_Name, Dept_ID, Dept_Name, Dept_Head)

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#### Decomposition into 3NF:

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1. **EMPLOYEE (Emp_ID, Emp_Name, Dept_ID)**

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## 2. Core Concepts & Memory Keywords

- **Transitive Dependency:** Non-key attribute determining another non-key attribute.

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## 3. Must-Write Points for Exams

- 3NF guarantees **lossless-join decomposition** and **dependency preservation**.

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4. Quick Recall Flow

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2NF R
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